

PARSNIP—NODE

Meshtastic LoRa mesh node, 915 MHz
Schematic

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REVISION HISTORY

REV	DATE	DESCRIPTION OF CHANGES	BY
1.1	2026—10—03	Prototype schematic; PCB layout in progress	LimesKey

NOTES, UNLESS OTHERWISE SPECIFIED:

- Resistance in ohms, capacitance in farads. All voltages are DC.
- Resistors and capacitors are 0402.
- DNP = do not populate. DNP symbols are drawn with a red X.
- An overbar on a pin or net name denotes an active—low signal.
- Labels with the same name are connected. Top—level sheets join through global labels and power symbols, subsheets of Root through sheet pins.
- PWR_FLAG symbols are for ERC only and are not parts.
- Some symbols are edited in the schematic. Do not run Tools > Update Symbols from Library.

LimesKey

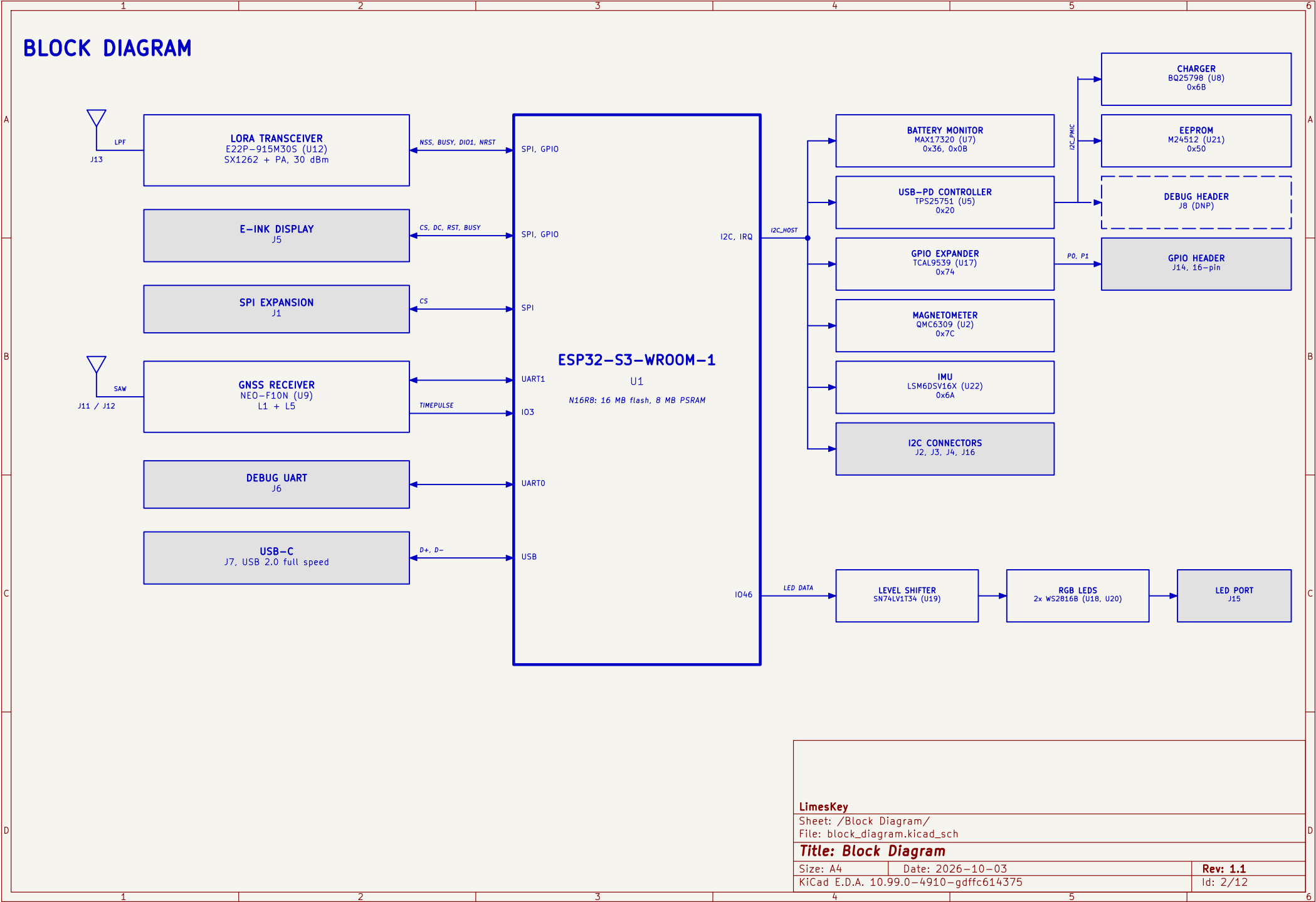
Sheet: /Overview/
File: overview.kicad_sch

Title: Cover

Size: A4 Date: 2026—10—03
KiCad E.D.A. 10.99.0—4910—gdffc614375

Rev: 1.1
Id: 1/12

BLOCK DIAGRAM



LimesKey

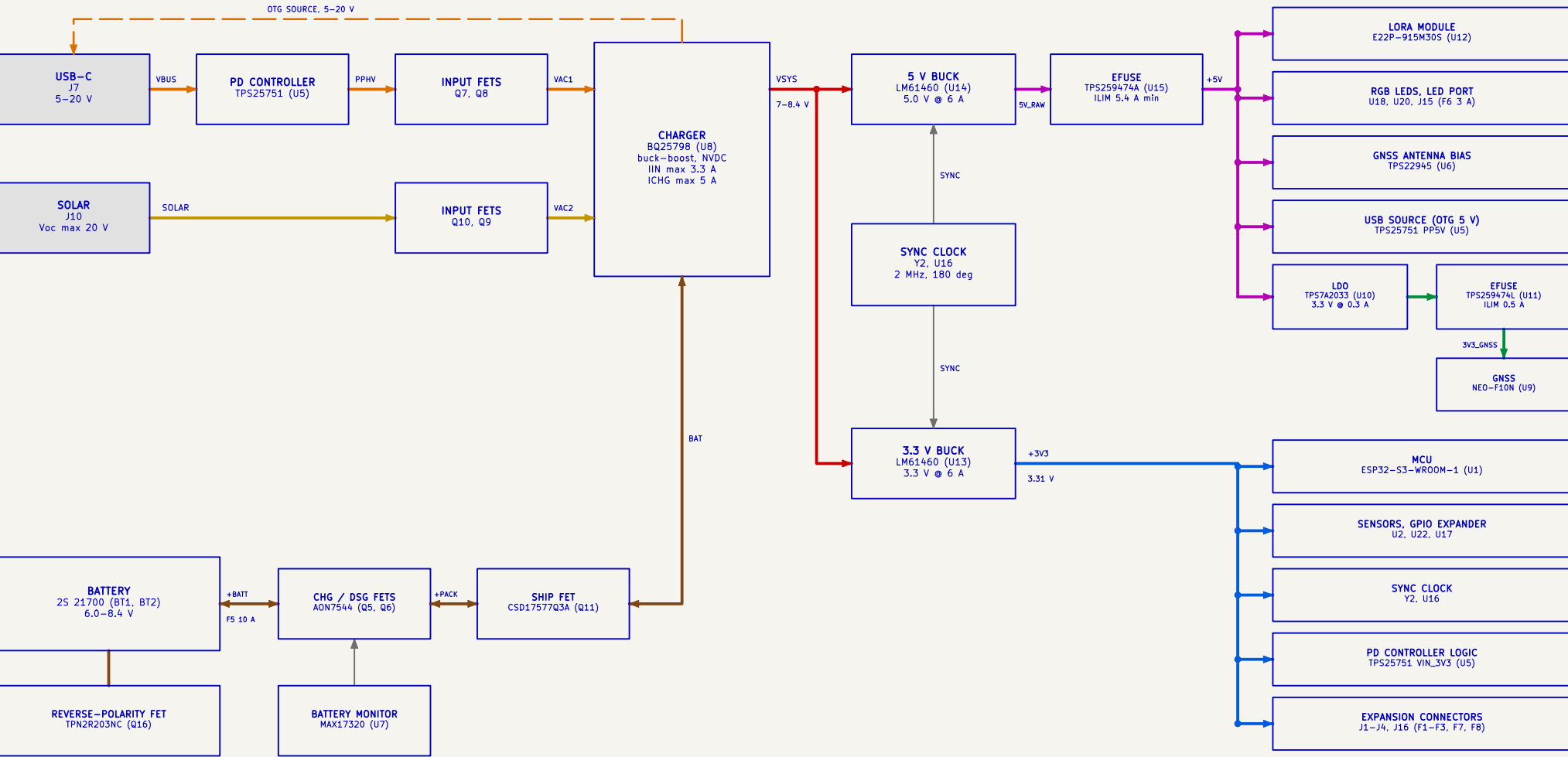
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File: block_diagram.kicad_sch

Title: Block Diagram

Size: A4 Date: 2026-10-03
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev: 1.1
Id: 2/12

POWER TREE



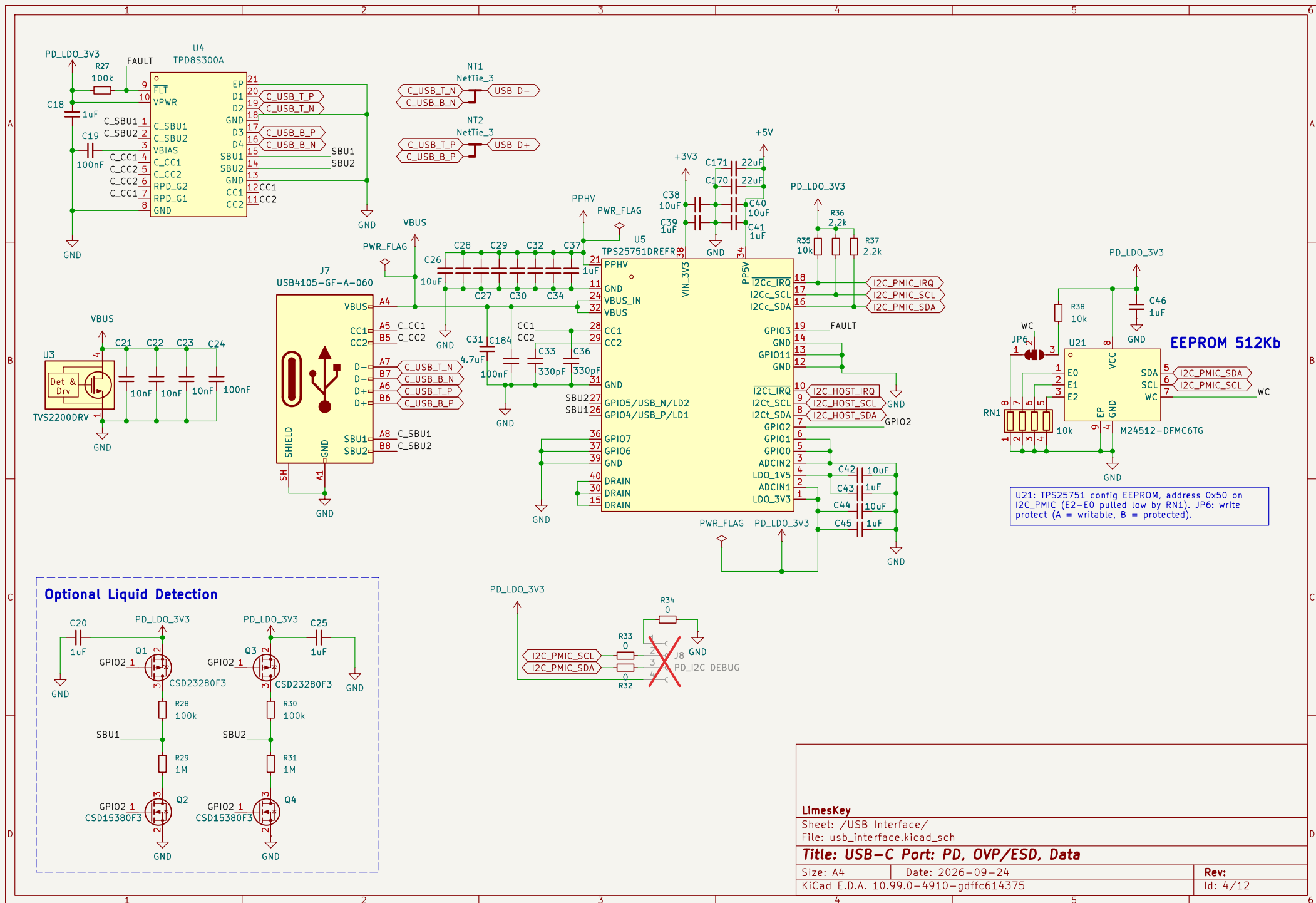
LimesKey

Sheet: /Power Tree/
File: power_tree.kicad_sch

Title: Power Tree

Size: A4
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev: 1.1
Id: 3/12



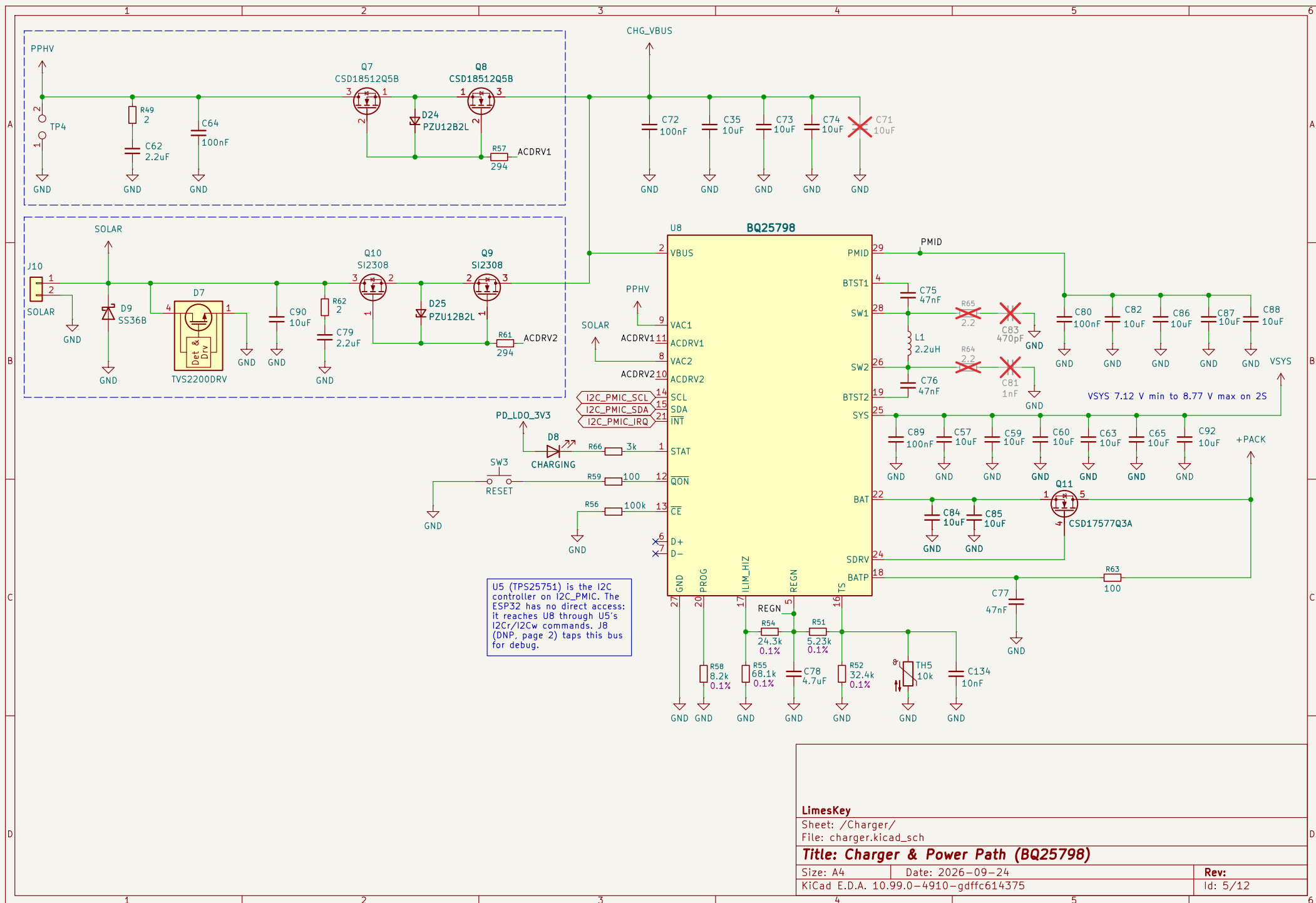
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Sheet: /USB Interface/
File: usb_interface.kicad_sch

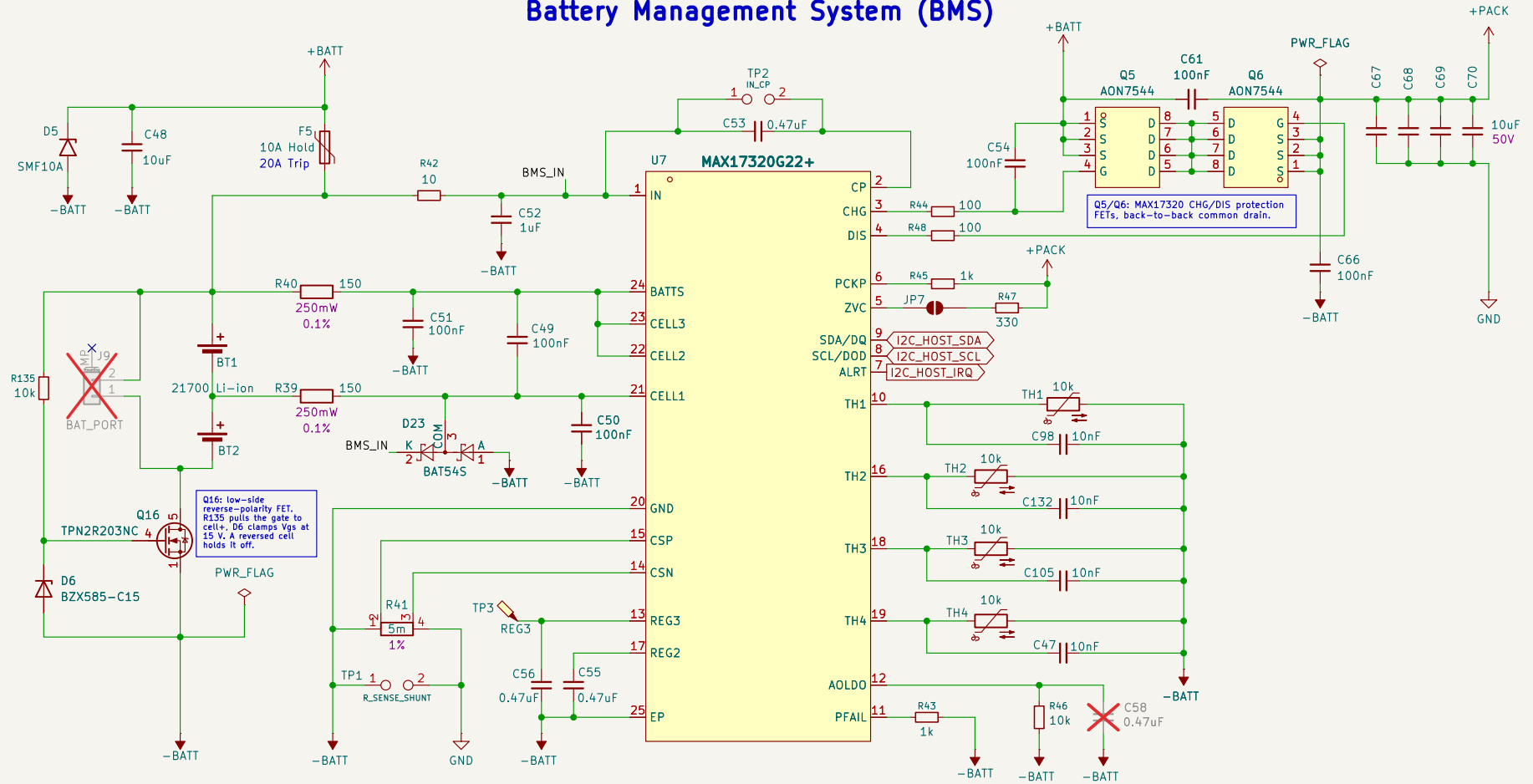
Title: USB-C Port: PD, OVP/ESD, Data

Size: A4 Date: 2026-09-24
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev:
Id: 4/12



Battery Management System (BMS)



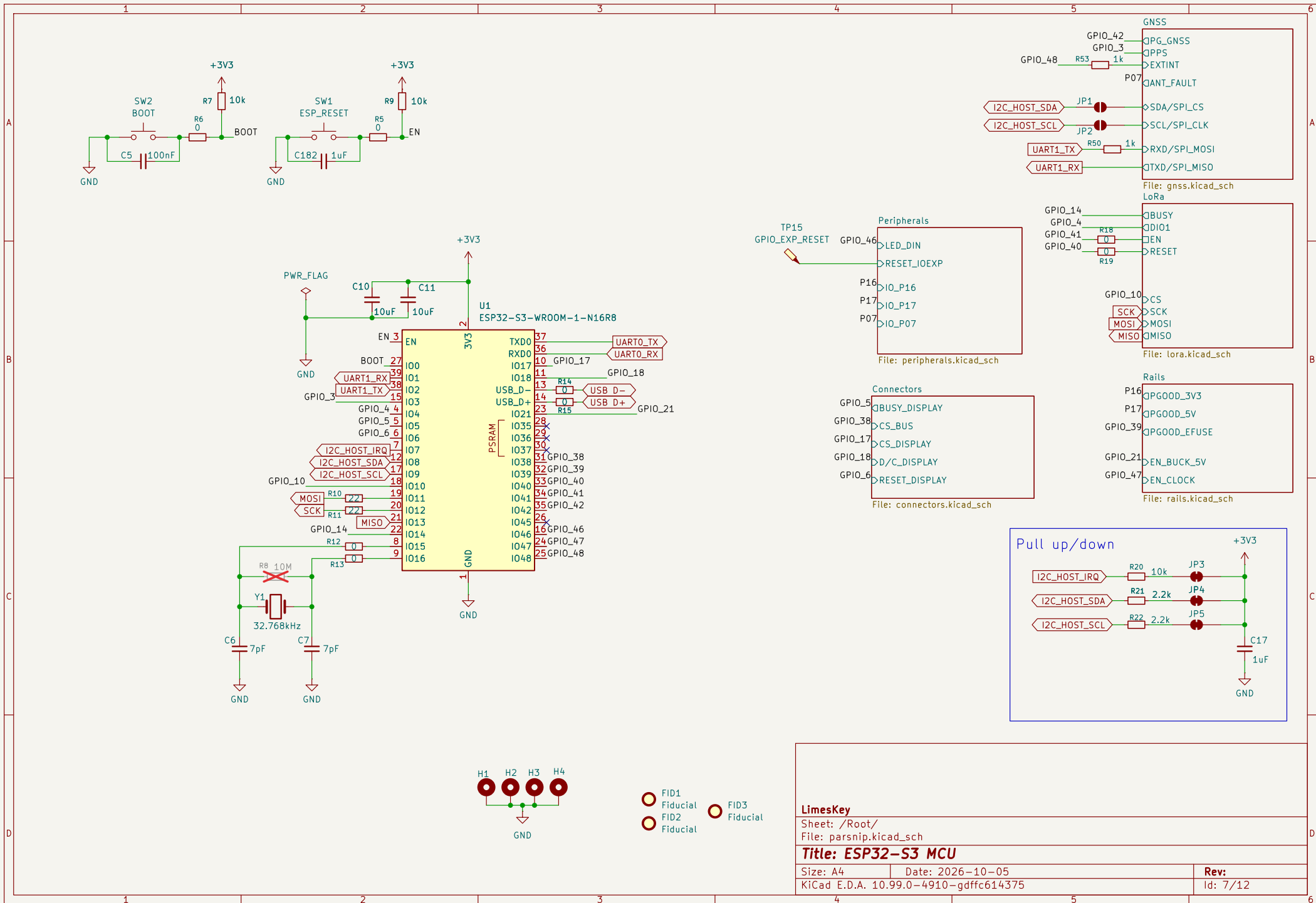
LimesKey

Sheet: /BMS/
File: bms.kicad_sch

Title: Battery Management System (BMS)

Size: A4 Date: 2026-09-27
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev:
Id: 6/12



LimesKey

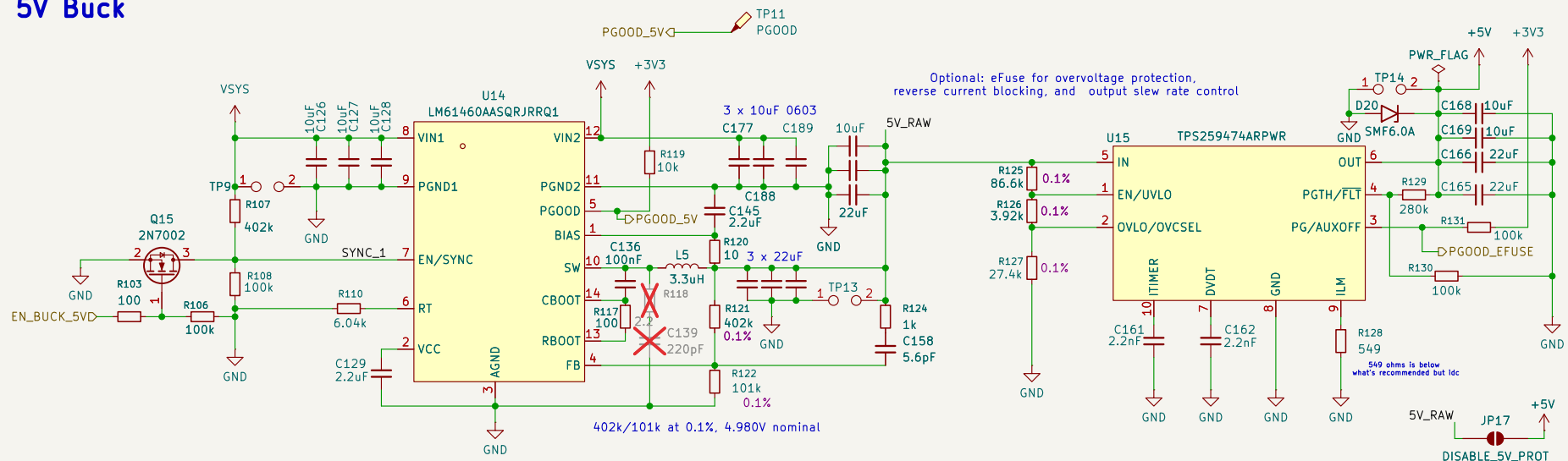
Sheet: /Root/
File: parsnip.kicad_sch

Title: ESP32-S3 MCU

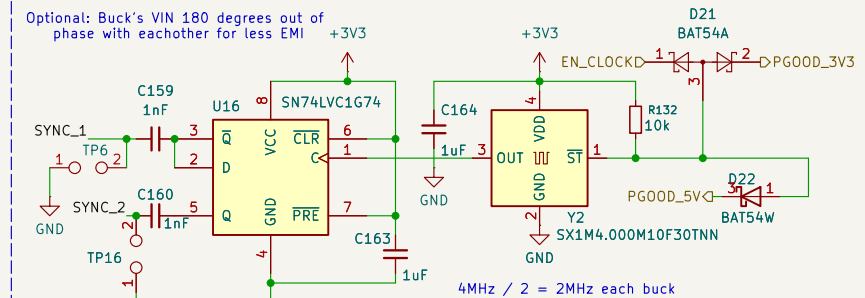
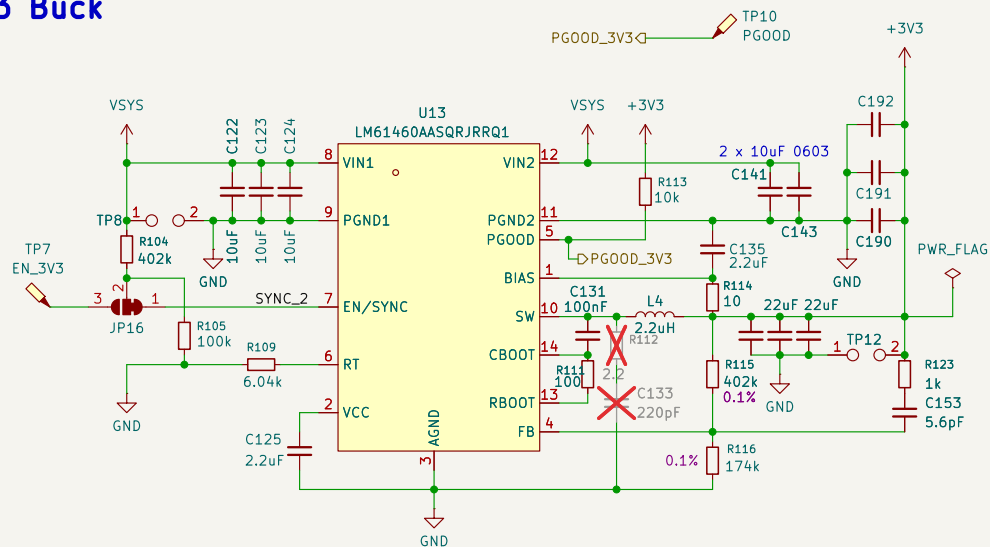
Size: A4 Date: 2026-10-05
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev:
Id: 7/12

5V Buck



3V3 Buck



LimesKey

Sheet: /Root/Rails/

File: rails.kicad_sch

Title: 5 V & 3.3 V Buck Regulators

Size: A4

Date: 2026-10-05

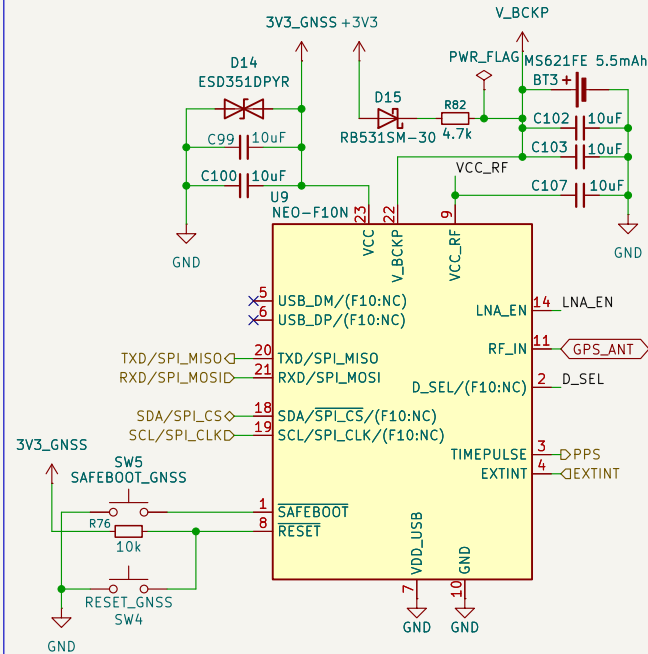
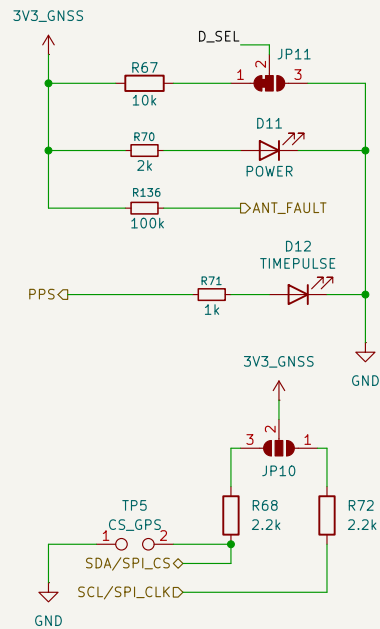
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev:

Id: 8/12

LEDs and Jumpers

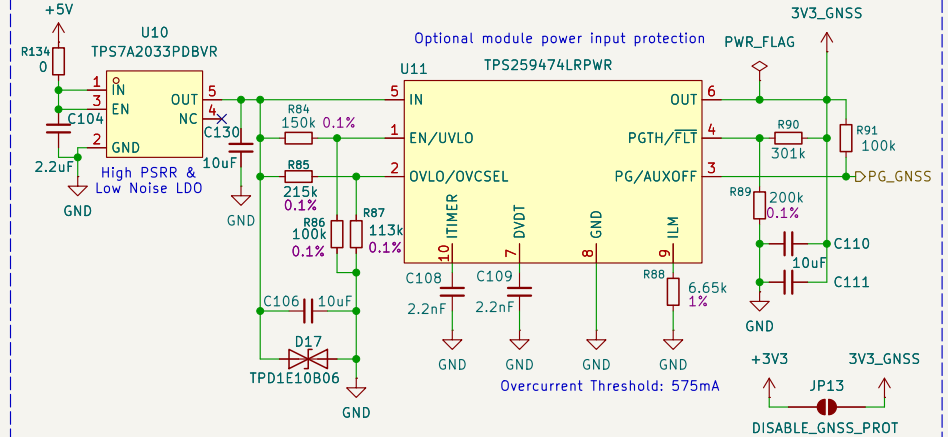
High to enable I2C/UART, low to enable SPI
Only for NEO-M[1]N



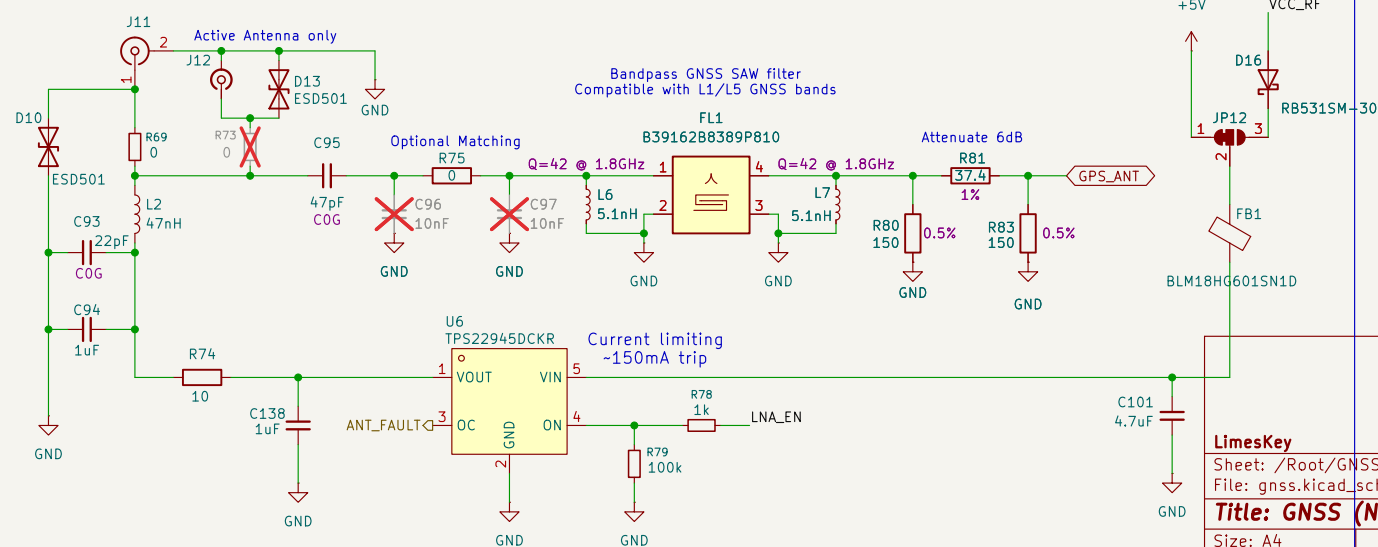
Pin-Pin compatible with NEO-M9N/M8N/M7N/M6N & NEO-F10N
The NEO-F10N only supports UART

Sparkfun Schematic: https://cdn.sparkfun.com/assets/1/0/a/8/a/SparkFun_GPS_NEO_M9N_SMA_UPDATE.pdf
U-Blox Datasheet: https://content.u-blox.com/sites/default/files/NEO-M9N_Integrationmanual_UBX-19014286.pdf

Optional LDO & eFuse



SMA + IPEX Connector



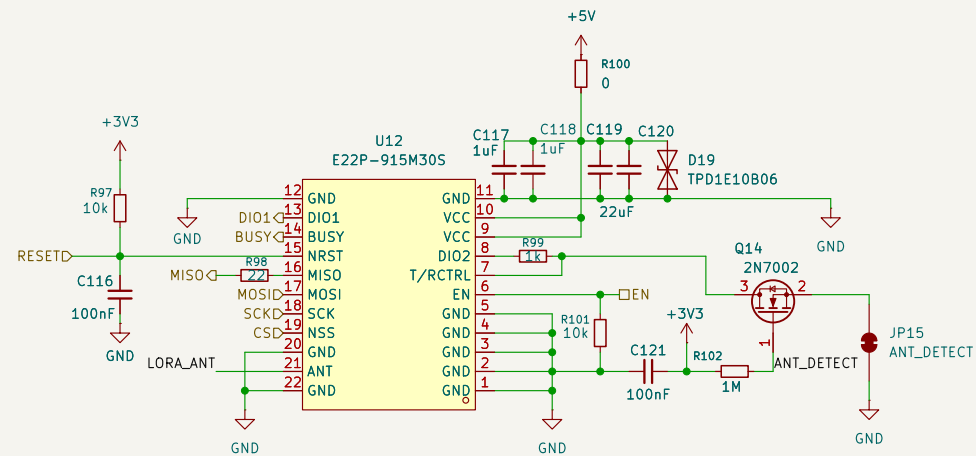
LimesKey

Sheet: /Root/GNSS/
File: gnss.kicad_sch

Title: GNSS (NEO-F10N) & Active Antenna

Size: A4 Date: 2026-09-14
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev: 1.1
Id: 9/12



SMA Connector

900MHz Wide Band Low Pass Filter

SMA-KE-347-H13.5-1.6

FL2 0900LP15B0063001E

DC Block C115

L3 470nH

R92 10k

D18 PGB1010603

C112 1nF

C113 1pF

C114 1pF

R95

JP14 BYPASS_FILTER

C115 47pF

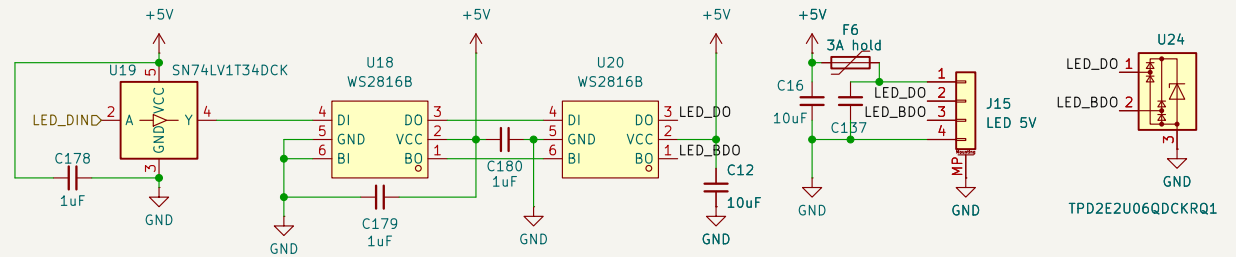
LORA_ANT

ANT_DETECT

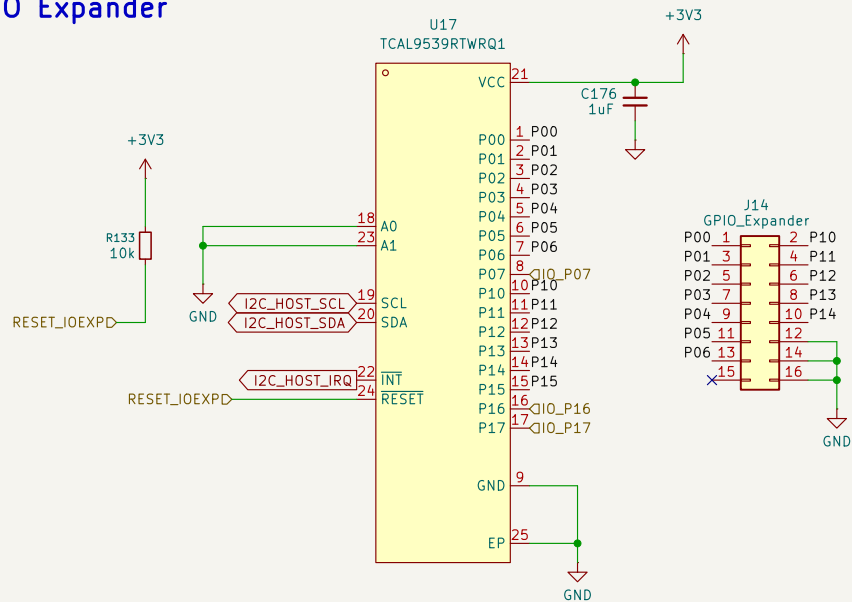
GND

Id: 10/12

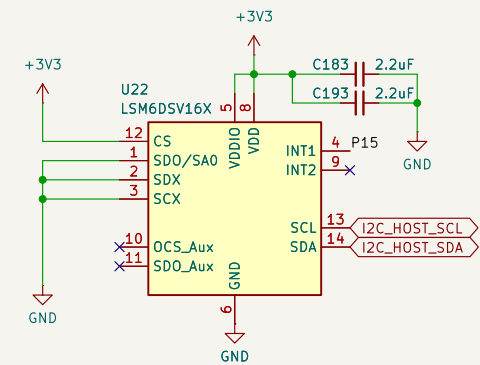
Individually Addressable RGB LEDs



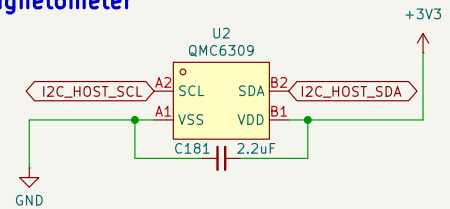
GPIO Expander



IMU



Magnetometer



LimesKey

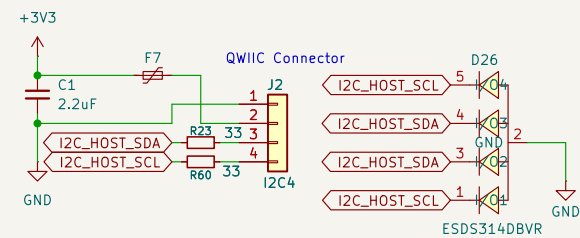
Sheet: /Root/Peripherals/
File: peripherals.kicad_sch

Title: GPIO Expander, RGB LEDs, Magnetometer

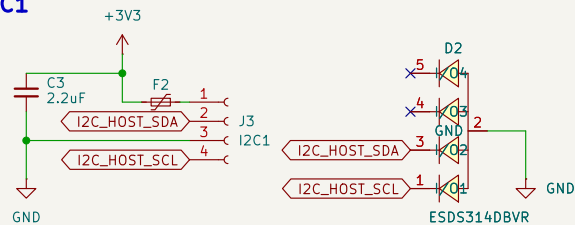
Size: A4	Date: 2026-10-05
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Rev:
Id: 11/12

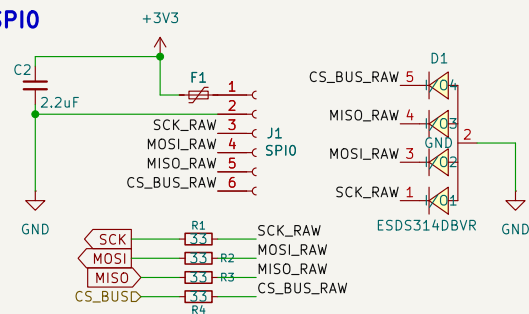
QWIIC I2C4



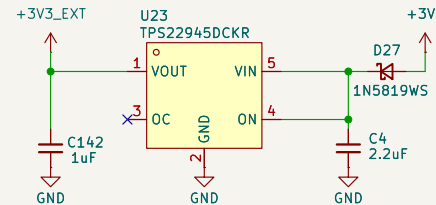
I2C1



SPI0

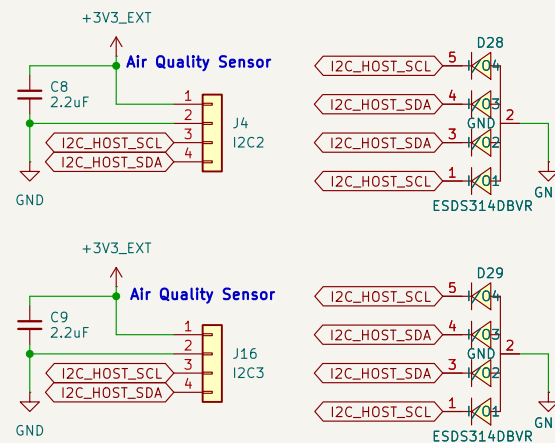


Low-input-voltage Current-limited Load Switches With Shut Off And Auto-Restart Feature

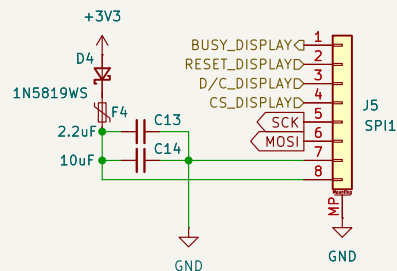


U23: Cin capacitance must be greater than Cout capacitance (locally)

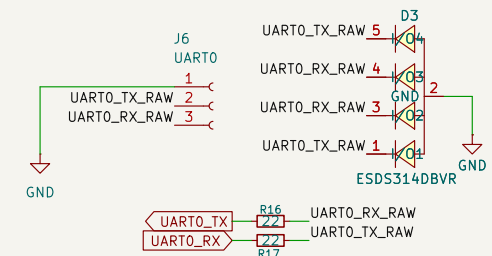
I2C2 & I2C3



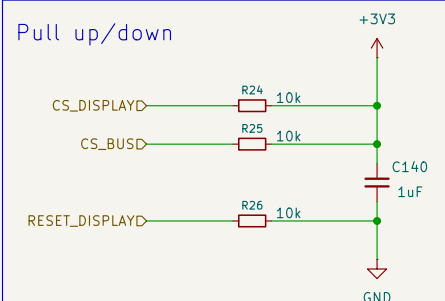
ePaper Display SPI1



UART0



Pull up/down



LimesKey

Sheet: /Root/Connectors/
File: connectors.kicad_sch

Title: Expansion & Display Connectors

Size: A4 Date: 2026-10-05
KiCad E.D.A. 10.99.0-4910-gdffc614375

Rev:
Id: 12/12